

The landscape of negative biases: modality, exclamative-questions and exclamatives

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Abstract

The interaction between modality and evidentiality has received considerable attention, but much less is known about the interaction between modality and speech-act operators. This paper argues that this interface is crucial to understanding how evaluative meanings arise with epistemic modals, particularly in non-information-seeking interrogatives and exclamatives. We develop an account of modal bias as a manipulable component of modal meaning and propose that question and exclamative operators can themselves act as bias manipulators. The empirical domain is Italian epistemic future, whose exceptional flexibility makes visible the mechanisms underlying this interaction. Building on previous work, we analyze the Italian future as a universal epistemic modal with no default bias, rather than as an evidential or a marker of subjective credence. Its flexibility derives from a metaevaluative layer at which the speaker's evaluation of the prejacent is encoded and can be modified.

The analysis distinguishes two mechanisms of negative bias: in *mica* future questions, bias emerges through the interaction of modality, QUES and the particle; in future exclamatives, it is introduced directly by EXCL. More generally, the paper argues that speech-act operators participate in semantic composition with modal meaning by manipulating its metaevaluative dimension, while remaining distinct from modality itself.

Keywords: modality, speech acts, negative biases, future, mica, interrogatives, exclamatives

While empirical and theoretical approaches have extensively explored the numerous connections between modality and evidentiality,¹ the interface between modality and speech act has been largely overlooked. We submit that this interface is key to understanding how evaluation with modals arises, most notably with non-information seeking interrogatives and exclamatives.

This paper addresses the interaction between epistemic modality and speech acts operators, by rethinking the notion of modal bias. Bias is understood as a ranking mechanism (Kratzer,

¹It would be impossible to render justice to the vast literature on these connections. For key contributions across different frameworks, see, among others, Karttunen (1971); Van der Auwera and Plungian (1998); De Haan (1999); Faller (2002); Matthewson and Frank (2005); Von Stechow and Gillies (2010); Aikhenvald and Dixon (2014); Giannakidou and Mari (2016), as well as reference works such as Portner (2009); Nuyts and Van Der Auwera (2016).

1991; Portner, 2009), which, as recently shown by Giannakidou and Mari (2021b), can be manipulated rather than serving as a fixed element of the lexical modal meaning. This paper theorizes that questions and exclamations can be bias manipulators and compose with the modal meaning.

Manipulations of biases are particularly salient with what the literature has dubbed ‘unitary modals’ Matthewson (2013), that is to say modals that are unspecified for force. In Gitksan, epistemic modality *imaa* can have either existential (1) or universal force (2).²

Context: Someone asks: "Whats that noise?"

- (1) *limx imaa/ima' t* Bob
sing EPIST DM Bob
Bob might be singing.

Context: Joe left the meeting looking really green in the face and sweaty. Someone asks you why he left.

- (2) *yugw imaa hl siipxw-t*
IMPF EPIST CN sick-3SG.II
He must have been sick.

Italian future is well-known to have an epistemic modal semantics akin to MUST³ and MIGHT, by notably being compatible with embedded PAST (4)-(6)-(8) and PRES/statives, (3)-(5)-(7).

- (3) Flavio sarà nella sua camera.
Flavio be.FUT.3SG in-the his room
Flavio must/might be in his room.'
- (4) Flavio avrà dormito nella sua camera.
Flavio have.FUT.3SG slept in-the his room
Flavio must/might have slept in his room.
- (5) Flavio deve essere nella sua camera.
Flavio must be in-the his room
Flavio must be in his room.
- (6) Flavio deve aver dormito nella sua camera.
Flavio must have slept in-the his room
Flavio must/might have slept in his room.
- (7) Flavio può essere nella sua camera.
Flavio might be in-the his room
Flavio might be in his room.
- (8) Flavio può aver dormito nella sua camera.
Flavio might have slept in-the his room
Flavio might have slept in his room.

However, unlike epistemic modals MIGHT and MUST that are specified for force, Italian future seems to very naturally align with unitary modals (1)-(2) – or flexible modals – as its ability to combine with adverbs of different strengths reveals.⁴

²Birman shows a similar pattern for circumstantial modality *ya*¹ that has both a possibility and a necessity reading. See Vittrant (2013) for extended discussion.

³We use capital letters for expressions type cross-linguistically, and italics for specific expressions of a target language.

⁴See for extended discussion Bertinetto (1979); Pietrandrea (2005); Mari (2009, 2010, 2021, 2024); Baranzini and Mari (2019); Giannakidou and Mari (2023).

- (9) a. Sarà sicuramente a casa.
 be.FUT certainly at home
 He must certainly be home.
 b. Sarà forse a casa.
 be.FUT maybe at home
 He might maybe be home.

This pattern has been explained in terms of underspecification (e.g. (Mari, 2010), following the approach of Matthewson (2013)).⁵ More recently, it has been analyzed in terms of *biases*, following the framework of Kratzer (1991): the strong interpretation, arising from the combination with a strong adverb (*sicuramente, certainly*), triggers a positive bias toward the truthfulness of proposition *p*, whereas the weak interpretation, arising from the combination with a weak adverb (*forse, maybe*), reflects a lack of bias. Moving forward Kratzer’s view, Giannakidou and Mari (2018a, 2021b) develop a theory according to which modals have a default force and a default bias (or lack thereof) but the default bias can be *manipulated*. Biases reveal the speaker’s confidence about the truthfulness of *p*, and are thus the mirror of an evaluative content. This content is encoded in a special position in the modal skeleton, called *meatevaluation* (Giannakidou and Mari, 2018a, 2021b). In this paper we build on these results and show that the speech act operators can act as a metaevaluators.⁶

With much of the literature on MUST, Giannakidou and Mari (2021b) focus on positive biases. The study of exclamatives highlights *negative biases*, namely *counter-expectation* and *mira-tive bias*. As we will spell out in great detail, these two biases highlight different ways in which the speaker can dissent: either by preferring worlds in which *p* is not true or by focusing on worlds in which stereotypical laws do not hold. These two biases are manifested, respectively, in exclamative questions with future and *mica* (10) and in exclamatives with future (11).

Context: it is snowing outside, and Flavio is not home.

- (10) Non sarà mica andato a navigare a Saint Quentin ?!
 not be.FUT.3SG MICA go.PAST.PART to sail at Saint-Quentin ?!
 He did not go to sail to Saint Quentin, isn’t he ?

Context: I find out that Flavio indeed went sailing with his coach while it is snowing.

- (11) Saranno fuori di testa !
 FUT out of mind !
 They are totally out of their mind !

These biases, as evaluations of propositional content, as unexpected or even surprising, arise under very specific evidential conditions. As a result, evidentiality, modality, and speech acts are intertwined in a way that this paper will attempt to clarify.

To achieve this, we will rely on previous work that has established that the Italian future tense is not an evidential but a necessity modal without bias (Mari, 2021; Giannakidou and Mari, 2023). We will immediately summarize the results of those previous works and then build on them to explain the ability of the future tense to convey contrariness biases.

The paper is structured as follows. We briefly introduce and justify the modal theory of Italian future in Section 1 (Mari, 2009; Giannakidou and Mari, 2018b, 2023). We recall the main arguments for not treating the Italian future as an evidential in Section 1.1, present the theory for Italian future as non-biased epistemic modal in section 1.2. In the discussion, we introduce the important notion of metaevaluation which will be key to understanding the flexibility of future in assertions, questions, and exclamatives. In Section 2 we discuss exclamative questions with *mica* and future in exclamatives and provide a new analysis for contrariness biases. Section

⁵For a discussion of this proposal, see Ippolito and Farkas (2021).

⁶See Mari (2021, 2024); Giannakidou and Mari (2026a) for reflective questions with Italian future.

3 concludes briefly.

We achieve two main results: (i) we spell out the analytical contributions of modality, QUES and the particles to the enhancement of an exclamative question and (ii) we provide an analysis for the modality/EXCL interaction that predicts mirativity.

Overall, we elaborate a new view according to which speech act operators compose with the modal meaning and the particles and contribute to the semantic interpretation of the utterances, without being reducible to modals themselves.⁷

1 Background on Italian future

1.1 The future is not an evidential

The intricacies between evidentiality and epistemic modality are both descriptive and theoretical. From an analytic point of view, it is controversial whether evidentiality is reducible to modality (Matthewson et al., 2007) or whether the two are to be analyzed at two quite distinct levels (Aikhenvald (2004), Faller (2002)). There is consensus that epistemic modals are sensitive to sources of knowledge, since the foundational work of Kratzer (1981). The literature advocates indirect (Karttunen, 1972), inferential (Von Fintel and Gillies, 2010) or partial evidence (Giannakidou and Mari, 2016) as presuppositions of the epistemic meaning of modals and we will align with this latter position here. We furthermore endorse the view that Italian future is a non-biased universal modal, that can be strengthened into a biased necessity modal (Mari, 2021; Giannakidou and Mari, 2023). This strengthening, we will recall, occurs at the level of the metaevaluation.

The presentation of these previous results proceeds in three steps. We first review the arguments in favor of an evidential analysis of the Italian future (Section 1.1), and show that they cannot settle the question. We also show that FUT is not a subjectivity marker conveying mere credence (*pace* Mari (2010) and Ippolito and Farkas (2021)). Section 1.2 presents our background analysis of Italian future as an epistemic modal, and introduces the empirical phenomenon of modal spread and metaevaluation as an analytical tool (Giannakidou and Mari, 2018a, 2021b).

1.1.1 The future is not an inferential evidential

The task of disentangling epistemic modality from evidentiality is a complex one that fails to provide tests that definitely settle the question.

The first test is the *ban against direct perception*.⁸ Italian future is certainly banned from direct evidence contexts (12) and it is felicitous in inferential contexts (13) (Mari, 2009; Giannakidou and Mari, 2016). However, this cannot be used as an argument for considering the Italian future as an inferential evidential. *Dovere* ('must') patterns just alike: it is infelicitous in direct evidence contexts (14) and felicitous in inferential evidence contexts (15) (see a.o. Karttunen (1972), Portner (2009), Von Fintel and Gillies (2010), Giannakidou and Mari (2016), Goodhue (2017)).

- (12) Context: Direct visual perception of rain, the speaker sees the rain falling.
#Starà piovendo.
It must be raining
- (13) Context: I see a wet umbrella.
Starà piovendo.
It must be raining

⁷Pace Kaufmann (2012) for imperatives, for instance.

⁸The ban against direct perception is one of the most debated arguments in the literature of both epistemic modality (Karttunen (1972), Portner (2009), Von Fintel and Gillies (2010), Goodhue (2017), a.o.) and Italian future (Mari (2009), Giannakidou and Mari (2016), Frana and Menéndez-Benito (2019) a.o.)

- (14) Context: Direct visual perception of rain, the speaker sees the rain falling.
#Deve star piovendo.
It must be raining
- (15) Context: I see a wet umbrella.
Deve star piovendo.
It must be raining

The second test used to advocate the evidential nature of Italian future is *challengeability*. It is correct that a future claim, just like evidential claims, cannot be challenged.

- (16) A: Why is Luca absent today ?
B: Sarà.FUT malato. – He might be ill.
A: That's not true. He is perfectly fine.
A: #That's not true. You saw he was ill.

It is well-known, however, that epistemic modals cannot be challenged either (see Papafragou (2006)).

- (17) A: Why is Luca absent today ?
B: Deve essere malato malato. – He must be ill.
A: That's not true. He is perfectly fine.
A: #That's not true. You saw he was ill.

Frana and Menéndez-Benito (2019) also advocates *speakers's orientation*. They argue that, unlike epistemic modals (18), the future cannot be anchored to the addressee (19).

- (18) Context: Robert's mom is running after the school bus with a snack. Robert is not on the bus. Lucie, who is with Robert, asks Robert why his mom is running after the bus. Robert replies (endorsing his mom's perspective):
I might be on the bus.
- (19) Context: same as in (18).
#Sarò sull'autobus.
I might be on the bus.

According to Frana and Menendez-Benito (*ibid.*), because the Italian future cannot be anchored to the addressee, unlike English *might*, Italian future is not a modal.

Again, in Italian, the future patterns just like the modal *potere* (*might/can*) in the indicative (20-b), which is also infelicitous in the same context. In Italian, anchoring to the addressee is possible with *potere* in the conditional (20-c).

- (20) (The felicity is relative to addressee anchoring)
- a. #Sarò sull' autobus.
 be.FUT.1SG on-the-bus
 - b. #Posso essere sull'autobus.
 might.IND.1SG be on-the-bus
 - c. Potrei essere sull'autobus.
 might.COND.1SG be on-the-bus

A final argument to plea for future as an evidential comes from its behavior in *questions*. It has been claimed that, in questions, Italian future triggers the interrogative flip (Frana and Menéndez-Benito, 2019; Ippolito and Farkas, 2019).

The interrogative flip is argued to be enhanced by evidentials in biased questions (Bhadra (2020), Korotkova (2016), Murray (2016), a.o.), and is the phenomenon whereby the source of

the evidence needed to answer the question is the addressee, unlike in assertives, whose source of evidence is the speaker.

The reasoning goes as follows: (i) a question containing an evidential expects that the hearer has the relevant type of evidence (interrogative flip e.g. Speas and Tenny (2003), Aikhenvald (2004), Murray (2016); San Roque et al. (2017), Bhadra (2020) a.o.) (ii) the evidential future is an inferential (iii) the future question is well-formed only if the hearer does not have direct evidence to answer the question. This elegantly explains the infelicity of (21) as an information seeking question.

- (21) #Quanti anni avrai ?
 how many years have.FUT.2SG
 How old might you be?

This test fails to tease apart evidentiality from epistemic modality. First, we note that the same explanation applies for theories holding that the future is an epistemic modal with an evidential presupposition, as the infelicity of (21) as an information seeking question would be a case of presupposition failure. Second, epistemic modals are infelicitous in information seeking questions as well, just like Italian future (and evidentials cross-linguistically).

- (22) #Quanti anni puoi avere ?
 how many years can.FUT.2SG have
 How old might you be?

The interrogative flip is an instance of a more general mechanism whereby, with questions, the addressee is given epistemic authority to answer the question Giannakidou and Mari (2026b). As a consequence, the knowledge relevant for answering is the knowledge of the addressee.

In virtue of being anti-knowledge markers, modals and Italian in future in questions, impose that the addressee does not have full knowledge to answer the question.

On the assumption that epistemic modals are anti-knowledge markers (Karttunen (1971); Kratzer (1991); Giannakidou and Mari (2016, 2018a); Lassiter (2017); Goodhue (2017)), the very fact that questions give the addressee epistemic authority when information seeking, explains that epistemic modals are infelicitous in context where the addressee knows the answers.⁹ Anti-knowledge is also visible in the compatibility of modals, and even strong modals like *must*, with 'I am not sure' or similar continuations as argued in Lassiter (2017).¹⁰

- (23) This is a very early, very correct Mustang that has been in a private collection for a long time. ... The speedo[meter] shows 38,000 miles and *it must be 138,000, but I don't know for sure.*
- (24) *I don't know for sure, sweetie, but she must have been very depressed. A person doesn't do something like that lightly.*
- (25) It must have been a Tuesday (*but I don't know for sure*), I can't remember"

To conclude, none of the tests seems to be able to disentangle evidentiality from epistemic modality and to show that the Italian future is not an epistemic modal, and all the facts discussed in this section can be accounted for on the assumption that the Italian future is an epistemic modal with a presupposition of indirect evidence (Pietrandrea, 2005; Mari, 2009; Giannakidou and Mari, 2016, 2018a).

⁹As the incompatibility with direct perception reveals, Italian future is indeed incompatible with full knowledge (Mari, 2009; Giannakidou and Mari, 2016), giannakidou mari 2018b, giannakidou mari 2023reply

¹⁰For a very extended discussion, see Giannakidou and Mari (2021b).

1.1.2 The future is not a subjective evidential

There is another understanding of the evidentiality of Italian future that has been repeatedly discussed in the literature (Squartini, 2010; Mari, 2010; Ippolito and Farkas, 2021), namely subjectivity in the sense of Nuyts (2001).

Evidence for the subjective evidential nature of the Italian future comes from its ability to be used in context where there is a *lack of factual evidence* (26-a). As highlighted in (26-b) and (26-c) neither *dovere* (*must*), nor *potere* (*might*) can be used in contexts where the speaker is in the dark and has no factual evidence at hand.

- (26) Where is Gianni ?
- a. Bo'. Sarà a casa !
bah! Be.FUT.3SG at home !
He might be home!
 - b. Bo'. #Deve essere a casa !
bah! Must be at home !
He must be home!
 - c. Bo'. #Può essere a casa !
bah! Might be at home !
He might be home!

The Italian future being a subjective evidential explains why it is weird for a doctor to use a sentence in the future: it is expected from a doctor that she uses factual evidence and not ungrounded subjective appreciations (or *endogenous* evidence in the sense of, a.o. Nuyts (1992); Mari (2016b); Giannakidou and Mari (2021a)).

Context: the doctor has visited John and utters:

- (27) Sarà il Covid.
be.FUT.3SG the Covid
It will be Covid.

The conclusion that FUT conveys pure credence either in its modal base, or in its ordering source seems to follow from these observations. Mari (2010) provided the first account in this direction. The core idea that Mari (*ibid.*, p. 1-8) advanced is that the Italian future uses 'subjective stereotypicality', that is to say, a set of speaker preferences, enhancing a ranking 'according to what it is more likely for the speaker.' Squartini (2010) also developed a view according to which the future relies on the speaker as the only epistemic authority. Finally, in a more recent paper Ippolito and Farkas (2019) defend the idea of 'subjective likelihood' of Italian future.¹¹

However, that it would be erroneous to still hold on to this view. Indeed, Italian future statements can also rely on *exogenous* evidence, as in (13), which we repeat here in (29). At most, only the ranking might be subjective: the modal base is epistemic and relies on factual evidence (the presence of a wet umbrella).

- (29) Context: I see a wet umbrella.
Starà piovendo.
It must be raining.

We propose that compatibility of the Italian future with lack of exogenous evidence is a mani-

¹¹The fact that FUT can be used without any clue, whereas MUST uses evidence, explains the most striking difference between the two: FUT is used to *project a world* that does not yet exist (temporal use of the future), MUST mandatorily uses clues and thus requires that the world already exists. It cannot therefore be future oriented and epistemic. There is not even an epistemic inference (*pace* Condoravdi (2002)).

- (28) John must come with us. (#Epistemic)

festation of its hallmark of extreme flexibility in the evidential presupposition and, as a consequence, its modal force.

1.2 Italian future as non-biased epistemic necessity

According to Mari (2021); Giannakidou and Mari (2023), Italian future is a flexible modal that can be very naturally strengthened or weakened, as the initial observation of its compatibility with adverbs of different forces reveals (30) (= (9)).

- (30) a. Sarà sicuramente a casa.
be.FUT.3SG certainly at home
He must certainly be home.
b. Sarà forse a casa.
be.FUT.3SG maybe at home
He might maybe be home.

In the following discussion, we provide the framework within which the notion of non-biased epistemic necessity is elaborated.

1.2.1 First and secondary modal bases without ordering

To outline the analysis, we will draw on the work of Giannakidou and Mari (2016, 2018b, 2021a,b),¹² highlighting only the key components. The foundational concept we will begin with is the epistemic state of an individual anchor i , which, in the case of the Italian future tense, always corresponds to the speaker.

- (31) *Epistemic state of an individual anchor i*
An epistemic state $M(i)$ is a set of worlds associated with an individual i representing worlds compatible with what i knows or believes in the context of utterance.

This is equivalent to Kratzer's epistemic modal basis (Kratzer, 1981, 1991), anchored to an epistemic agent i .

A nonveridical state conveys epistemic uncertainty if (and only if) it is partitioned into p and $\neg p$ worlds.

- (32) *Nonveridical information state*
An information state $M(i)$ is nonveridical about p iff $M(i)$ contains both p and $\neg p$ worlds.

The future tense (FUT) employs a primary epistemic modal base, $M(i)$, which reflects the assumption that the speaker's lack of knowledge arises from partial or corrupted evidence, or even from a complete absence of evidence, leaving the speaker in a state of uncertainty (Giannakidou and Mari, 2016).

Ideal_S delivers the worlds in the modal base in which all the propositions in S are true.

- (33) $\text{Ideal}_S M(i) = \{w' \in M(i) : \forall q \in S (w' \in q)\}$

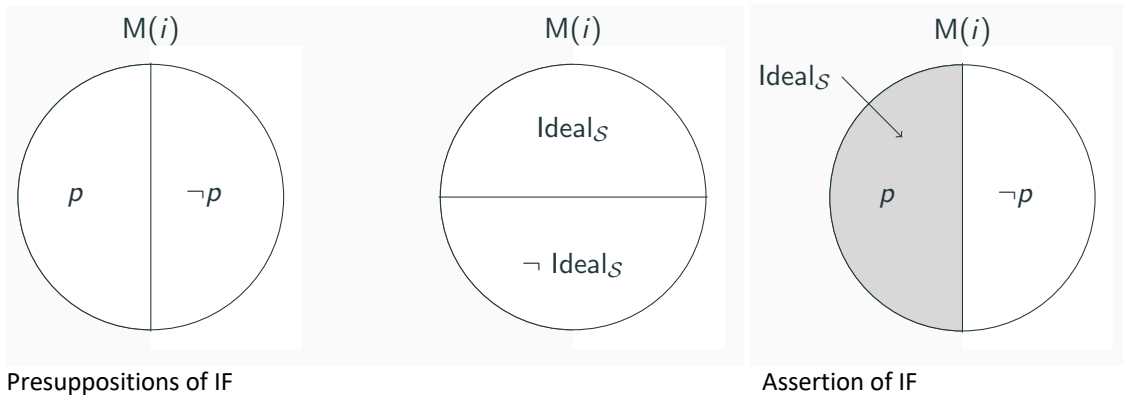
S is a set of heterogeneous propositions that correspond to common ground norms/personal but also personal views etc. Ideal_S can encode i 's beliefs, but not necessarily only mere credences. One's beliefs can also rely on considerations of stereotypicality conditions or normalcy conditions. Note that Ideal_S is a secondary modal base and does not provide ranking: Ideal_S worlds are not ranked higher than Ideal_S ones (Mari, 2021; Giannakidou and Mari, 2023).

¹²All definitions in this Section are from Giannakidou and Mari (2021b).

1.2.2 Italian future (similarities and differences with MUST)

The non-final lexical entry for Italian future (FUT) is in (34) and picture 4 illustrates it. We see there that the modal base $M(i)$ is partitioned between p and $\neg p$ worlds, as well as between Ideal_S and $\neg \text{Ideal}_S$ worlds. FUT asserts that all Ideal_S worlds are p worlds. Note that the presupposition also encodes evidential bias, expanding on the previous analysis of Giannakidou and Mari (2023).

- (34) (non-final) $\llbracket \text{FUT (PRES (} p \text{))} \rrbracket^{M, \text{Ideal}, i, S}$ is defined only if (i) $M(i)$ is nonveridical and (ii) $M(i)$ is partitioned into Ideal_S and $\neg \text{Ideal}_S$ worlds, and (iii) i **has potentially no clues**.
 If defined,
 $\llbracket \text{FUT (PRES (} p \text{))} \rrbracket^{M, \text{Ideal}, i, S} = 1$ iff $\forall w' \in \text{Ideal}_S : p(w', t_u)$
Paraphrase: (Presupposition) The modal base $M(i)$ is partitioned into p and $\neg p$ worlds as well as into Ideal_S and $\neg \text{Ideal}_S$ worlds. i has potentially no clues for p .
 (Assertion). All Ideal worlds are p worlds.



FUT and MUST are both universal epistemic modals that quantify over Ideal_S worlds (Giannakidou and Mari, 2016).

- (35) (non final) $\llbracket \text{MUST (PRES (} p \text{))} \rrbracket^{\text{Epist}, \text{Ideal}, i, S}$ is defined only if (i) $M(i)$ is nonveridical and (ii) $M(i)$ is partitioned into Ideal_S and $\neg \text{Ideal}_S$ worlds, and (iii) i **has some reliable evidence**.
 If defined,
 $\llbracket \text{MUST (PRES (} p \text{))} \rrbracket^{\text{Epist}, \text{Ideal}, i, S} = 1$ iff $\forall w' \in \text{Ideal}_S : p(w', t_u)$
Paraphrase: (presupposition) The modal base Epist is partitioned into p and $\neg p$ worlds as well as into Ideal_S and $\neg \text{Ideal}_S$ worlds. i has some reliable evidence for p . (assertion). All Ideal worlds are p worlds.

The difference lies in the type of evidence that FUT and MUST rely upon: FUT is compatible with potentially lack of evidence (26-a); MUST requires that the speaker has some reliable evidence (26-b).

This *evidential bias* (Giannakidou and Mari, 2021a), in turn, determines the speaker's evaluation of the truthfulness of the prejacent. With FUT the speaker might not be biased towards p , whereas with MUST positive bias is the default. The bias (or lack thereof) is encoded in the metaevaluation as we now explain.

1.3 Modal spread

The initial characterization put forth by Lyons (1977) described modal verbs and auxiliaries as capable of combining with modal adverbs whose force is comparable of that of the modal itself (see also Huitink (2012), Willer (2013)). According to this view, MUST can combine with

adverbs like DEFINITELY or PROBABLY (36), while MIGHT can combine with adverbs like MAYBE (37).

(36) John must definitely/probably be at home.

(37) He might possibly be at home.

However, all modals exhibit a cross-linguistic tendency toward flexibility, being compatible with a range of adverbs that express varying degrees of confidence on the part of the speaker. Giannakidou and Mari (2018a) refers to this phenomenon as modal spread and demonstrates that it is subject to cross-linguistic variation. In Italian (as in English), we observe mismatches where a weak modal combines with a stronger adverb (38)-(39), and a strong modal combines with a weaker adverb (40)-(41). They provide the following examples to illustrate the phenomenon.

(38) e a questa circostanza **può probabilmente** essere dovuto il fatto che egli fosse arrivato al nono compleanno. Comunque stessero le cose, in ogni modo, era il suo nono compleanno.

and the fact that he reached his ninth birthday might probably be due to these circumstances. No matter how the facts where settled, in any case, it was his ninth birthday.¹³

(39) In some cases, however, the psychosis **might definitely** be due to anxieties and conflicts associated with the pregnancy.¹⁴

(40) Il vaso, che costituisce uno dei premi guadagnati dagli atleti negli agoni panatenaici di Atene, **deve forse** fare parte del corredo di una sepoltura ubicata non lontano dall'area di Castel Nuovo.

The jar, which constitutes one of the prizes earned by the athletes in the pan-athenians olympics of Athens, must maybe belong to the kid of a burial located not far from the area of Castel Nuovo.¹⁵

(41) So there **must maybe** be some glitch somewhere along the line or something that makes this happen. I am sure is a cache or technical glitchup.¹⁶

Giannakidou and Mari (2018a) propose that the modal skeleton features an obligatory metaevaluator, which can be overt (or $\emptyset$), and which, by default, has a force equal to the one of the modal.

We interpret the Italian future's ability to combine with adverbs of varying strengths, illustrated with some attested examples, as a manifestation of modal spread.

(42) **Sarà sicuramente** a casa.
be.FUT.3SG certainly at home
He must certainly be home.

(43) **Sarà sicuramente** stato contento che tante persone ... abbiano espresso per lui una grande stima ...
he must have been pleased that so many people ... have expressed high esteem for him.¹⁷

(44) **Sarà forse** a casa
be.FUT.3SG maybe at home
He might maybe be home.

¹³<https://books.google.fr/books?isbn=8804536829>

¹⁴<https://books.google.com/books?id=c6JPyf0BZYIC&pg=PA74&lpg=PA74&dq=%22might+definitely>

¹⁵<http://www.comune.napoli.it/flex/cm/pages/ServeBLOB.php/L/IT/IDPagina/1425/UT/systemPrint>

¹⁶<https://www.blackhatworld.com/seo/ogads-com-mobile-cpa-cpi-incent-network-mobile-content-locker-high-cr-704909/page-26>.

¹⁷<http://albo.comunefinaleligure.it/web/>

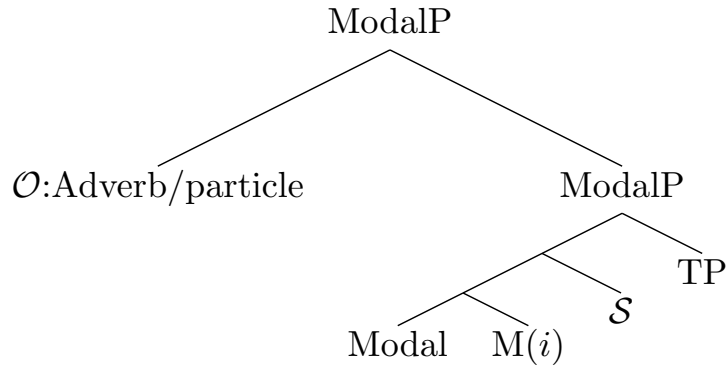


Figure 1: Modal skeleton with metaevaluation Giannakidou and Mari (2018a)

- (45) Sarà.FUT la magia di quei luoghi ... **Sarà forse** che Siena è speciale, unica. Fatto sta che non c'è stato un solo singolo giorno in cui non mi sia sentito a casa ...
 'It might be the magic of those places ... It **might be maybe** be that Siena is special, unique. The fact is that there was not a single day when I did not feel at home ...'¹⁸

Giannakidou and Mari (2018a) propose that when adverbs (or any other bias manipulator, including particles like *mica*, as we will discuss) combine with modals, they either maintain or alter the default bias (or lack of bias) of the modal. Giannakidou and Mari (2018a) provide the following definitions:

- (46) Effect of the adverbs with universal modals.
- a. DEFINITELY (It. *assolutamente*; Gk. *oposdhipote*; Eng. *definitely*): Strengthening the default positive bias.
 - b. PROBABLY (It. *probabilmente*; Gk. *mallon*; Eng. *probably*): Maintaining the default.
 - c. MAYBE (It. *forse*; Gk. *isos*; Eng. *maybe*): Weakening the default.
- (47) Effect of the adverbs with existential modals.
- a. DEFINITELY (It. *assolutamente*; Gk. *oposdhipote*; Eng. *definitely*): Introducing positive bias.
 - b. PROBABLY (It. *probabilmente*; Gk. *mallon*; Eng. *probably*): Introducing positive bias.
 - c. MAYBE (It. *forse*; Gk. *isos*; Eng. *maybe*): Maintaining the default.

To account for the composition, a silent adverb is posited for any modal, supporting the idea that each modal has a default bias at the metaevaluation layer.

MUST has a default bias of weak necessity Kratzer (1991), and with this extra layer of metaevaluation, the final lexical entry for MUST is the following (expanded from Giannakidou and Mari (2018a)).

- (48) $\llbracket \emptyset \text{ MUST (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S}$ is defined only if (i) the modal base $M(i)$ is nonveridical, (ii) it is partitioned into Ideal_S and $\neg \text{Ideal}_S$ worlds, and (iii) i **has some reliable evidence**.

If defined,

$\llbracket \emptyset \text{ MUST (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S} = 1$ iff: Ideal_S is a weak necessity with respect to $\neg \text{Ideal}_S$ relative to $M(i)$ and $\mathcal{O}$ & $\forall w' \in \text{Ideal}_S : p(w', t_u)$

Paraphrase: (presupposition) The modal base Epist is partitioned into p and $\neg p$ worlds

¹⁸<https://www.brntolodielasua.it/sara-forse-che-siena-e-speciale-unica>

as well as into Ideal_S and $\neg\text{Ideal}_S$ worlds. i has some reliable evidence for p . (assertion). Ideal worlds are weak necessity and they are p worlds.

This default is maintained when MUST is combined with PROBABLY.

- (49) $\llbracket \text{PROBABLY MUST (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S}$ is defined only if (i) the modal base $M(i)$ is non-veridical, (ii) it is partitioned into Ideal_S and $\neg\text{Ideal}_S$ worlds, and (iii) i **has some reliable evidence**.
If defined,
 $\llbracket \text{PROBABLY MUST (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S} = 1$ iff: Ideal_S is a weak necessity with respect to $\neg\text{Ideal}_S$ relative to $M(i)$ and $\mathcal{O}$ & $\forall w' \in \text{Ideal}_S : p(w', t_u)$

The weak necessity default of MUST can be strengthened by DEFINITELY into necessity Kratzer (1991).

- (50) $\llbracket \text{DEFINITELY MUST (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S}$ is defined only if (i) the modal base $M(i)$ is non-veridical, (ii) it is partitioned into Ideal_S and $\neg\text{Ideal}_S$ worlds, and (iii) i **has some reliable evidence**.
If defined,
 $\llbracket \text{DEFINITELY MUST (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S} = 1$ iff: Ideal_S is a necessity with respect to $\neg\text{Ideal}_S$ relative to $M(i)$ and $\mathcal{O}$ & $\forall w' \in \text{Ideal}_S : p(w', t_u)$

Unlike for Kratzer (1991) for whom the ordering source determines both a secondary restriction and an ordering, for Giannakidou and Mari (2018a), the secondary restriction via Ideal_S and the ordering are two independent parameters. This independence allows manipulations that account for several observations, and most notably the modal flexibility.

The default for Italian future is lack of bias, and the resulting final lexical entry will be as follows (expanding on Giannakidou and Mari (2023)).

- (51) $\llbracket \emptyset \text{ FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S}$ is defined only if (i) $M(i)$ is nonveridical and (ii) $M(i)$ is partitioned into Ideal_S and $\neg\text{Ideal}_S$ worlds, and (iii) i **has potentially no clues**.
If defined,
 $\llbracket \emptyset \text{ FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S} = 1$ iff $\mathcal{O}$ is empty & $\forall w' \in \text{Ideal}_S : p(w', t_u)$
Paraphrase: (Presupposition) The modal base $M(i)$ is partitioned into p and $\neg p$ worlds as well as into Ideal_S and $\neg\text{Ideal}_S$ worlds. i has potentially no clues for p .
(Assertion). The metaevaluation is empty, and all Ideal worlds are p worlds.

The adverb MAYBE can be overt and routinely combine with FUT as illustrated in (44).¹⁹

- (52) $\llbracket \text{MAYBE FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S}$ is defined only if (i) $M(i)$ is nonveridical and (ii) $M(i)$ is partitioned into Ideal_S and $\neg\text{Ideal}_S$ worlds, and (iii) i **has potentially no clues**.
If defined,
 $\llbracket \text{MAYBE FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S} = 1$ iff $\mathcal{O}$ is empty & $\forall w' \in \text{Ideal}_S : p(w', t_u)$

Positive bias of Italian future can be strengthened by adverbs of the DEFINITELY type, as illustrated in (42) and implemented in (53).

- (53) $\llbracket \text{DEFINITELY FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S}$ is defined only if (i) $M(i)$ is nonveridical and (ii) $M(i)$ is partitioned into Ideal_S and $\neg\text{Ideal}_S$ worlds, and (iii) i **has potentially no clues**.
If defined,
 $\llbracket \text{DEFINITELY FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S} = 1$ iff
 Ideal_S is a necessity with respect to $\neg\text{Ideal}_S$ relative to $M(i)$ and $\mathcal{O}$ & $\forall w' \in \text{Ideal}_S : p(w', t_u)$

¹⁹The combination of MAYBE with MUST is more restricted, even in Italian, see discussion in Giannakidou and Mari (2018a).

FUT is a highly flexible modal, compatible with both a lack of bias and a positive bias. This flexibility stems from the evidential presupposition, which allows the assertion made with the future tense to be grounded in either factual and reliable evidence or a complete lack of clues.

This flexibility is further enhanced for Italian future in exclamative-questions and exclamatives, where negative biases arise. The landscape of these biases is heterogeneous, and we identify two configurations in which these arises: one in which $\neg p$ is the preferred alternative and one in which the set $\neg \text{Ideal}_S$ is quantified in.

2 Negative biases

2.1 Counterexpectation bias : MICA in future questions

This first type of contrariness bias that we study here is the counterexpectation bias that emerges when Italian future is combined with MICA.

- (54) Non sarai mica andato a dirgli il nostro segreto ?!
 Not be.FUT.2SG MICA went to tell him the our secret ?!
 You didn't go tell him our secret, isn't it ?

In normal circumstances it is expected by the speaker that the addressee hasn't revealed their secret. However, in (54) the speaker seems to possess contrary evidence to this expectation that warrants asking the question with a mirative stance.

Understanding of this complex interpretation requires to first delineate the semantic profile of MICA.

According to Cinque (1991), MICA can deny a proposition that has never been expressed linguistically but is instead inferred or assumed to be believed by the addressee. Cinque discusses a context in which the speaker S observes the addressee A pouring milk into a dish for S's cat. S infers that p (in this case: the milk is good for cats) is assumed by A.

- (55) Guarda che il latte non fa mica bene al gatto.
 look that the milk NEG does MICA good to-the cat
 Look, milk is NOT good for cats !

According to Cinque's analysis, NON introduces a regular negation, while MICA adds a definedness condition: in the context c , p is presupposed by some participant in the conversation.

Frana and Rawlins (2019) provide new evidence against Cinque's analysis and elaborate a scenario in which Clara invites Miles for dinner, making it clear to him that she will prepare her best dishes. When he arrives, Miles barely touches any food. Clara asks him:

- (56) Non hai mica già mangiato?
 not have.PRES.2SG MICA already eaten?
 You haven't eaten, don't you ?

Clara expects that Miles has *not* eaten yet; she is biased towards $\neg p$ but has evidence (evidence bias) that p (Miles barely touches any food).

Frana and Rawlins (2019) maintain Cinque's generalization for assertions, but advance a different generalization for questions, overall proposing what follows:

- (57) Frana and Rawlins (2019) generalization.
- a. In **assertions**, MICA sentences are felicitous in contexts where p is assumed by some participant in the conversation (as in Cinque (1991)).
 - b. **Questions** with MICA are felicitous if the speaker has prior bias (prior to the exchange) for $\neg p$.

We show that this generalization is not correct for questions, and that a unified account for both assertions and questions is possible.

We observe that, even in questions, p can be expected to be part of the common ground (rather not $\neg p$), as the following case, where p is expected to be true by the speaker, shows. The context is one in which I am eating a big steak and my friend asks me :

- (58) Scusa, non sei mica a vegetariana ?
 sorry, not are MICA vegetarian ?
 Sorry to ask, aren't you vegetarian ?

The speaker expects p to be true (that I am vegetarian) and has evidence for $\neg p$ (that I am eating a whole steak). We conclude that MICA is not sensitive to the valence of the speaker's bias presupposition.

We newly propose that MICA is anaphoric to rules and generalizations of the form (59), where E is a body of evidence (a set of propositions) available in the context of utterance c .

- (59) E then *normally* q

(59) is a defeasible inference that can be based on stereotypes, norms, commonsense reasoning. It is enhanced by contextual information or general knowledge (see a.o. Dowty (1979); König (1986, 1991); Krifka et al. (1995); Mari et al. (2012)). If we restate this rule in the framework that we have been designing, we obtain what follows:

- (60) MICA Non-at-issue.
 $\forall w' \in Ideal_S p(w')$

MICA states that a certain generalization does not hold and that the inference associated with the generic generalization should not go through, according to the speaker. In this sense, it is a contrariness element along the lines of Cinque (1991) (see discussion in Wu (2025)). Unlike Cinque, however, we propose that MICA presupposes a defeasible generalization.

- (61) MICA At-issue.
 $\forall w' \in Ideal_S \neg p(w')$

Unfolding (58), the reasoning goes as follows. Let the $Ideal_S$ worlds be those worlds in which I am eating a steak and $\neg p$ worlds, worlds in which I am not vegetarian.

- (62) a. $\forall w' \in Ideal_S \neg p(w')$ *defeasible generalization*
 b. $\forall w' \in Ideal_S p(w')$ *MICA content*
 c. p and $\neg p$ are valid alternatives in view of $Ideal_S$. *question*

In creating a question that is grounded in $Ideal_S$, the speaker is ready to endorse the contextual evidence, and nonetheless refuses to endorse $\neg p$. MICA turns out to be a disconfirmational particle, that rejects contextual evidence as valid (see also Aikhenvald (2004)).²⁰

Consider a scenario where the addressee (A) is far away, and the speaker (S) announces that the addressee's sister will pick him up (Cinque (1991); Frana and Rawlins (2019)). In making this announcement, the speaker appears to presuppose that the addressee's sister is capable of driving. However, the addressee clarifies that his sister is, in fact, unable to drive (63).

- (63) S: Sei lontano. Ti viene a prendere tua sorella.
 You are far away. Your sister is going to pick you up.
 A: Non sà mica guidare.
 She is not MICA able to drive.

²⁰Irrelevance of the evidence is also a semantic component essential to the concessive reasoning (König, 1986). On concessive future and the irrelevance of the evidence, see Baranzini and Mari (2019).

According to our account, what is crucial for MICA is the presence of a contextually available generalization that states: if someone picks up a person who is far away, they will likely use a car. While acknowledging and endorsing this generalization as evidence, MICA asserts that in all evidence-based worlds, $\neg p$ is also true.

2.2 Building counter-expectation bias with MICA

To account for the effect of MICA in questions, we start with the interpretation of future questions, on the assumption that QUES scopes over FUT.²¹ QUES creates two alternatives. One in which all Ideal_S are p worlds, and one in which only some of the Ideal_S are p worlds (see Figure 3).

- (64) $\llbracket \text{QUES (FUT (p))} \rrbracket^{\mathcal{O}, M, i, S}$ is defined only if:
 $M(i)$ is nonveridical and is partitioned into Ideal_S and $\neg \text{Ideal}_S$ worlds.
 $\llbracket \text{QUES (FUT (p))} \rrbracket^{\mathcal{O}, M, i, S} =$
 $\{ \{ \forall w' \in \text{Ideal}_S : p(w', t_u) \}, \{ \neg \forall w' \in \text{Ideal}_S : p(w', t_u) \} \}$

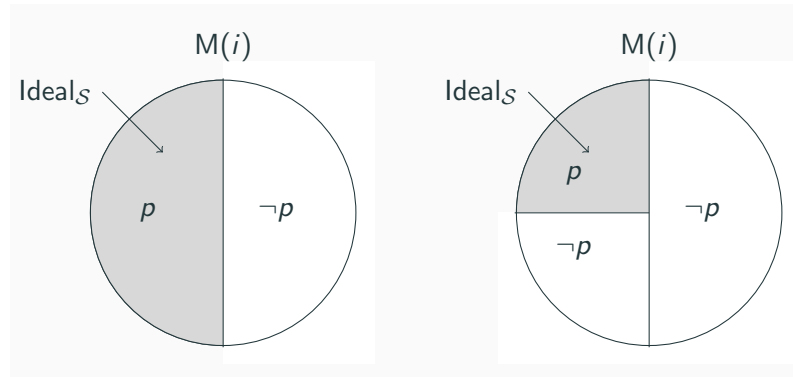


Figure 2: Non optimal question QUES > FUT – with uninterpreted MICA

This question is not ‘optimal’. Rephrasing it, it asks whether in all worlds compatible with Ideal_S p is true or whether there are only some of the worlds in Ideal_S in which p is true.

There are various reasons why this question is not well-formed, depending on the theory adopted. According to Hamblin (1958), Groenendijk and Stokhof (1982) and Mascarenhas (2009), questions denote sets of mutually disjoint alternatives. The alternatives pictured in Figure 2 are not disjoint.

According to a different view, any question opens up three options, which correspond to the possible answers *yes*, *no* and *I do not know* (Ciardelli et al., 2009; Enguehard, 2021).

Along these lines, we propose that MICA adds the alternative necessary to form a well-formed question, which we depict in Figure 3. The first alternative corresponds to the affirmative answer (all the Ideal_S worlds are p worlds), the second alternative to the ‘scenario answer’ (the Ideal_S set is partitioned), and the third alternative to the negative answer (all the Ideal_S worlds are $\neg p$ worlds).

The alternative contributed by MICA is interpreted at the metaevaluation level and is added to the interpretation as a third disjunct, completing the structure of a well-formed question (see Figure 3).

- (65) $\llbracket \text{MICA QUES (FUT (p))} \rrbracket^{\mathcal{O}, M, i, S} = \{ \forall w' \in \text{Ideal}_S : \neg p(w', t_u), \{ \forall w' \in \text{Ideal}_S : p(w', t_u) \}, \{ \neg \forall w' \in \text{Ideal}_S : p(w', t_u) \} \}$

²¹FUT can also scope over QUES, enhancing the ‘reflective’ interpretation (21). For extended discussion, see Mari (2024); Giannakidou and Mari (2026b).

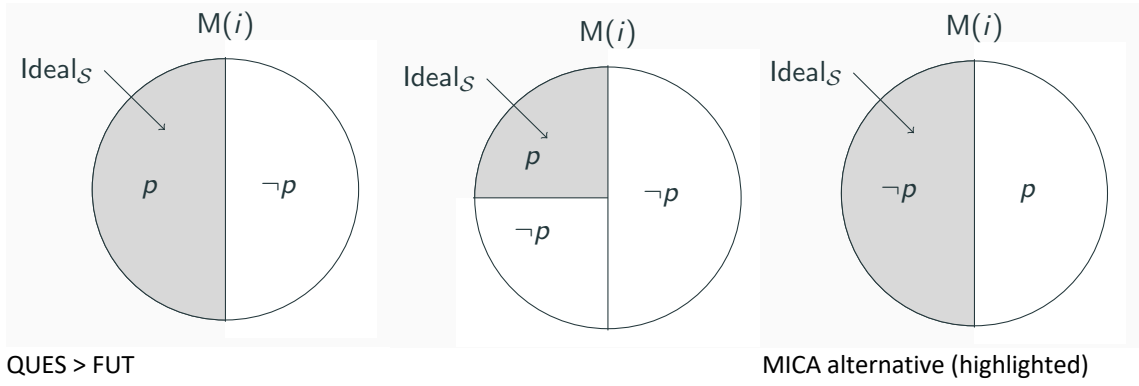


Figure 3: MICA future exclamative question

By being appended to the metaevaluation layer, where subjective evaluation is interpreted, the status of the MICA-alternative is distinct from the two other alternatives. This has two main consequences: first, an exclamative question is formed, characterized by a strong subjective component; second, the MICA-alternative is emphasized in this unique position, resulting in a biased **exclamative question**. Indeed, QUES is an alternative generator, and MICA introduces an alternative that reveals the speaker stance towards the evidence (as irrelevant). QUES and MICA thus share the task of creating an exclamative question.

Note that this addition is feasible because both the future tense and MICA are sensitive to a body of information available in the context of the utterance ($Ideal_S$). Inferences are drawn from $Ideal_S$.

Before concluding on MICA future questions, it is important to highlight a constraint on the co-occurrence of FUT and MICA in questions. Recall the contrast between the vegetarian case and the Miles-is-not-eating case. In the vegetarian case (66), the defeasible generalization undone by MICA leads to the conclusion that p . In contrast, in the Miles-is-not-eating case, the defeasible generalization results in the conclusion that $\neg p$. Additionally, note the use of the indicative present in (66) versus the future tense in (67).

- (66) Scusa, non sei mica vegetariana ? (= (58))
 Sorry, not are MICA vegetarian ?
 Sorry to ask, aren't you vegetarian ?
 Previous belief that p evidence that $\neg p$
- (67) Non avrai mica già mangiato? (= (56))
 not have.FUT.2SG MICA already eaten?
 You haven't eaten yet, don't you ?
 Previous belief that $\neg p$ evidence that p

The future can only be used if the contextual evidence leads to the conclusion that p , in particular, it cannot be used in (69) under Scenario 1.

Scenario 1: The speaker has contextual evidence that the addressee is *not* vegetarian and she is eating a whole steak.

- (68) Scusa, non #sarai mica vegetariana ?
 sorry, not be.FUT.2SG MICA vegetarian ?
 Sorry to ask, aren't you vegetarian ?

Consider Scenario 2. In this context, the future can be used.

Scenario 2: The speaker has evidence that the addressee *is* vegetarian. Indeed, there is a steak on the table and the addressee refrains from eating it.

- (69) Scusa, non sarai mica vegetariana ?
sorry, not are MICA vegetarian ?
Sorry to ask, aren't you vegetarian ?

FUT and MICA can co-occur only in the cases where the evidence supports p .

2.2.1 Conclusion on MICA future questions: evaluativity, rescuing and the formation of exclamative questions

The content that MICA introduces is inherently evaluative: by defeating a generalization with MICA, the speaker endorses a view that does not cope with what it is generally expected. Furthermore, this content is optional, as MICA is not obligatory and NON (plus prosody²²) suffices to enhance counter-expectation.

If our analysis is correct, the use of MICA in future questions highlights that evaluative content can contribute to the calculation of the meaning of the utterance. In the case of questions, this content is essential to rescue an otherwise ill-formed question.

This adjustment is possible insofar MICA and the modal share the same presupposition (evidence bias), and the alternative introduced by MICA, in virtue of its peculiar status, is highlighted and thus reveals the speaker's bias towards $\neg p$.

We have thus shown that the interaction between modality, QUES and MICA creates an exclamative question, a semantic object that is complex to apprehend, and for which the framework of the metaevaluation of modality offers useful analytical handles.

Finally, as a side note on biased questions and MICA, we observe that MICA questions with indicative present or imperfective are the equivalent of negative biased questions à la Ladd (see Ladd (1981) and subsequent literature).

- (70) Wasn't there a vegetarian restaurant around here ?

In Italian a negative question is ambiguous between a high and a low interpretation of the negation. MICA triggers the former reading and (71) is the Italian equivalent of (70).

- (71) Non c'era MICA un ristorante vegetariano nei dintorni ?

This question presupposes a previous positive expectation and conveys a negative stance towards p in view of contextual evidence. In the absence of the modal, and under the assumption that QUES also triggers a metaevaluative layer, MICA acts on the negative option that it ranks as higher. The illustration of the mechanism will deserve its own development elsewhere.

2.3 Mirative bias

2.3.1 Background on mirative futures in Spanish and Italian

Mirative future has been the focus of recent attention, most notably in the literature on Spanish future (a.o. Rosique (2015); Rodríguez Rosique (2019); Rivero (2014); Escandell-Vidal and Leonetti (2021)).

According to Rodríguez-Rosique, the mirative interpretation of the future tense arises from its inherent temporal "distance" component, which evolves into an expression of evaluative

²²The interpretation depicted in Figure 3 can be triggered by prosody in the absence of MICA. The study of its role and meaning in future questions is a matter that we leave to the specialists of the semantic prosody interface and move on to future exclamatives.

distance. When using the mirative future, the speaker does not fully endorse the truthfulness of *p*.

(72) (= (11) in Rosique (2015))

Y se había ido a escape a contárselo a ella, a su hermana Carmen: — ¡Será idiota el tonto que tengo por marido!

And he had rushed out to tell her, her sister Carmen: — Such an idiot, the stupid that I have as a husband!

(Francisco Blanes García, *El cura de Carboneras*. Entrelíneas Editores, 2009, p. 95).

According to Escandell-Vidal (2021), the Spanish future tense is both an inferential and a subjective element, and it maintains its core interpretation across various uses, including the mirative. Escandell-Vidal and Leonetti (2021) further argue that the primary illocutionary point of the mirative future is expressive. They state: “The speaker expresses their negative attitude toward the extreme degree to which a property is observed in a subject. Neither the propositional content nor the extreme degree is asserted; both are presupposed. The main illocutionary point is expressive. Surprise, extreme degree, and negative evaluation are, therefore, the key components of the mirative interpretation.”²³

To our knowledge Squartini (2018) is the only work that addresses mirative future in exclamatives in Italian. Importantly, a connection is made with epistemicity along the lines of Plungian (2010). Squartini argues that the epistemic nature associated with indirect information may be what triggers mirativity. Knowledge obtained indirectly—whether through inference or secondhand sources—is inherently less certain and thus carries an epistemic bias. This epistemic bias, as revealed in the mirative case, is ultimately tight to the speaker as the only authoritative source of knowledge. Across its variety of uses, future in Italian is, according to Squartini (see also Squartini (2012)), an inherently subjectivity marker.

(75) Sarà stanco !

be.FUT.3SG tired !

(Intended) He certainly is tired !

Having shown in section 1.1 that the Italian future cannot be a mere marker of subjectivity, we pursue a new analysis that builds on the epistemic nature of Italian future. We rely on the interaction between the metaevaluative layer of modality and EXCL to build the mirative interpretation.

In our account, the evaluative (mirative) component is substantiated as a ranking of less stereotypical worlds as higher and does not escape composition.

²³The label ‘mirative future’ is also used by Rivero (2014) for information that is not necessarily accepted by the speaker.

(73) Estaría histérica, pero le dije una verdad como un templo (ex. (32b) in Rivero (2014))

be.COND hysterical, but him told.1SG a truth like a temple

I might have been hysterical, but I told him a real big truth.

Here the speaker may not take responsibility for the content of the prejacent, which comes from another source, leading to situations where the denial of the prejacent is not excluded in principle. Importantly, the mirative use is attributed by Rivero to the conditional and not to the future itself. Furthermore, it is unclear that the mirative uses are those targeted here and also discussed by Rosique (2015) and Escandell-Vidal and Leonetti (2021). The constructions upon which Rivero comments are what Baranzini and Mari (2019) consider ‘concessive interpretations’. Interestingly enough, in Italian, the concessive interpretation arises with the future and not with the conditional, as in Spanish. As we have previously mentioned, there is a close connection between concessivity and mirativity, which we do not discuss here.

(74) Sarò stupida, ma mi sono laureata con il massimo dei voti.

be.FUT.1SG stupid but me am graduated with the maximum of the grades

I might be stupid, but I graduated with honors.

In the following sections, we empirically characterize future exclamatives and then provide an analysis in the metaevaluation framework.

2.3.2 Characterizing mirative future in exclamatives

Prima facie future exclamatives share several features with non-future (or bare) exclamatives.

Consider a context where my son and his coach left to sail while it is snowing outside. I exclaimate:

- (76) a. Saranno fuori di testa !
 be.FUT.3PL out of mind !
 They sure are out of their mind !
 b. Sono fuori di testa !
 be.PRES.3PL out of mind !
 They are out of their mind !

Both (76-a)-(76-b) satisfy the recency restriction (Rett and Murray, 2013): I newly learn that the boys went sailing while it is snowing outside and come to the conclusion that they are out of their mind. For both (76-a)-(76-b) the content is gradable and situated at the endpoint of a scale (Zanuttini and Portner, 2003; Rett, 2008, 2011) (scale of irresponsibility). Both (76-a)-(76-b) convey emotive expressive content (I am scared that something bad can happen). (Villalba, 2004, 2008; Castroviejo-Miró, 2008; Trotzke and Giannakidou, 2024).

Closer scrutiny, however, highlights some important differences between the bare exclamative (77-b) and the future exclamative (77-a).

Consider the following context. I am used to have my sons wash their hair every day. A friend of mine exclaimates:²⁴

- (77) a. Saranno puliti !
 be.FUT.3PL clean
 They sure are clean !
 b. Sono puliti !
 be.PRES.1PL puliti
 They are clean !

We take for granted that washing one's hair every day is certainly exaggerated and not stereotypically normal.

While both the future and the bare exclamatives reveal un-commitment of the speaker with respect to evidence (washing one's hair every day), the content in the future exclamative *follows* from the evidence, whereas the content in the bare exclamative *can precede* the learning of the information conveyed by the evidence.

The bare exclamative is felicitous if its content (the hair is clean) is considered to justify the inference that I should stop washing my son's hair and it is thus a fact (Grimshaw, 1979; Gutiérrez-Rexach, 2001; Zanuttini and Portner, 2003; Abels, 2010). The future exclamative presents *p* as a conjecture and maintains its inferential evidential presupposition that the Italian future has in assertions.

In Spanish this property is manifested through the frequent use of consecutive markers (we thank a reviewer for this note), where the equivalent of (75) is introduced by the consecutive *pues*:

- (78) ¡Pues estarán limpiísimos!

²⁴*Hair* is plural in Italian.

Furthermore, future maintains its hallmark of anti-factivity in exclamatives, This, in passing, corroborating Castroviejo-Miró (2008)’s view according to whom, the factivity of the exclamatives is a cancelable implicature rather than a presupposition.²⁵ This anti-factivity property is evident in the paraphrases of future exclamatives containing a disjunction (see (79-a)), mirroring the behavior observed in questions (see (79-b)).

Consider the following context: There is a weird man walking up and down in front of my fort door all day long. My partner exclaims:

- (79) a. Avrà una casa, una fidanzata ... !
Doesn’t he have a home **or** a girlfriend !
b. Ha una casa, una fidanzata, ... ?
Does he have a home **or** a girlfriend ?

Bare exclamatives (see (80-a)), on the other hand, align with bare assertions (see (80-b)) as they can be paraphrased using a conjunction.

- (80) a. Ha una casa, una fidanzata ... !
He has a home (**and**) a girlfriend !
b. Ha una casa, una fidanzata, ... He has a home (**and**) a girlfriend !

Note that (79-a) is an exclamation rather than a question or a modal assertion, as impossibility of the addition of *chi lo sa* (*who knows*) reveals.

- (81) a. Avrà una casa, una fidanzata ... ! #chi lo sa
Doesn’t he have a home **or** a girlfriend ! #Who knows
b. Avrà una casa, una fidanzata ... ? chi lo sa
Does he have a home **or** a girlfriend ? Who knows
c. Ha una casa, una fidanzata, ... ? Chi lo sa
Does he have a home **or** a girlfriend ? Who knows

The exclamative nature is also enhanced by *pure* (at least), as an overt minimizer.

- (82) a. Avrà **pure** una casa, una fidanzata ... !
Doesn’t he have at least a home **or** a girlfriend !

Nonetheless, the exclamative (beyond being anti-factive Zanuttini and Portner (2003))²⁶ is also imbued with uncertainty and conveys a sense of confusion or puzzlement, in agreement with the meaning of the future across different speech acts.

Overall the similarities and differences between bare exclamatives, future exclamatives and future assertions can be summarized as follows.

	Future	Future exclamative	Exclamative
Inference from the evidence	X	X	
Uncertainty	X	X	
Evidence is contradictory of previous expectations		X	X
p is presupposed			(X)

Table 1: Characterization of future exclamatives

²⁵Note, in passing, that (77-a) shows that the content does not need to be negative, unlike in Spanish (Escandell-Vidal and Leonetti, 2021) (see Squartini (2018) for the initial observation).

²⁶The term anti-factive is used by Zanuttini and Portner (2003) for verbs like *believe* or *think*, which do not always convey uncertainty. See discussion in Mari (2016a); Giannakidou and Mari (2021b); Mari and Portner (ress) for a distinction between strong and weak belief as revealed by mood variation. Uncertainty (in belief predicates and modality) is represented as a partitioned state, as per (32).

We now propose an analysis that explains how the interaction between future and EXCL produces mirativity.

2.4 Future exclamatives and mirative bias: analysis

Our proposal is that the exclamative introduces an evaluation that is interpreted in the metaevaluation. The exclamative ranks higher the non-Ideal_S worlds with respect to the Ideal_S ones.



Recall from (51) (repeated here in (84)) that Italian future is a modal that is born with a silent MAYBE:

- (84) $\llbracket \emptyset \text{ FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S}$ is defined only if (i) $M(i)$ is nonveridical and (ii) $M(i)$ is partitioned into Ideal_S and $\neg$ Ideal_S worlds, and (iii) i **has potentially no clues**.
 If defined,
 $\llbracket \emptyset \text{ FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S} = 1$ iff $\mathcal{O}$ is empty & $\forall w' \in \text{Ideal}_S : p(w', t_u)$
Paraphrase: (Presupposition) The modal base $M(i)$ is partitioned into p and $\neg p$ worlds as well as into Ideal_S and $\neg$ Ideal_S worlds. i has potentially no clues for p .
 (Assertion). The metaevaluation is empty, and all Ideal worlds are p worlds.

EXCL ranks $\neg$ Ideal_S as weak necessity, thus introducing negative bias.

- (85) $\llbracket \text{EXCL FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S}$ is defined only if (i) the modal base $M(i)$ is nonveridical, (ii) it is partitioned into Ideal_S and $\neg$ Ideal_S worlds, (iii) i **has potentially no clues**.
 If defined,
 $\llbracket \text{EXCL FUT (PRES } (p)) \rrbracket^{\mathcal{O}, M, i, S} = 1$ iff: $\neg$ Ideal_S is a weak necessity with respect to Ideal_S relative to $M(i)$ and $\mathcal{O}$ & $\forall w' \in \neg \text{Ideal}_S : p(w', t_u)$
Paraphrase: (Presupposition) The modal base $M(i)$ is partitioned into p and $\neg p$ worlds as well as into Ideal_S and $\neg$ Ideal_S worlds. i has potentially no clues for p .
 (Assertion). **Non-Ideal worlds are weak necessity and all non-Ideal worlds are p worlds.**

The analysis adheres to the hypothesis that the semantics of the future tense remains consistent across its various uses in different illocutionary environments, including assertions, exclamative questions, and exclamations. In Italian, the future tense has a modal meaning that presupposes uncertainty. The future conveys a conjecture that is based on evidence available in the context of the utterance. Specifically, in the case of future exclamatives, as highlighted in (85), the contextual evidence clashes with stereotypicality or normalcy conditions: the worlds of quantification are those that are the least stereotypical. This analysis preserves both the speaker's uncertainty and the anti-factivity inherent to the exclamative future.

Furthermore, the account proposed here aligns with the idea that exclamatives expand the domain of quantification to include non-stereotypical worlds (Zanuttini and Portner, 2003). However, in this case, there is no literal "widening." Instead, the EXCL operator selects $\neg$ Ideal_S worlds, ranking them higher than the stereotypical ones.

Overall, the analysis accounts for the observed distributions by leveraging the inherent flexibility of the Italian future tense.

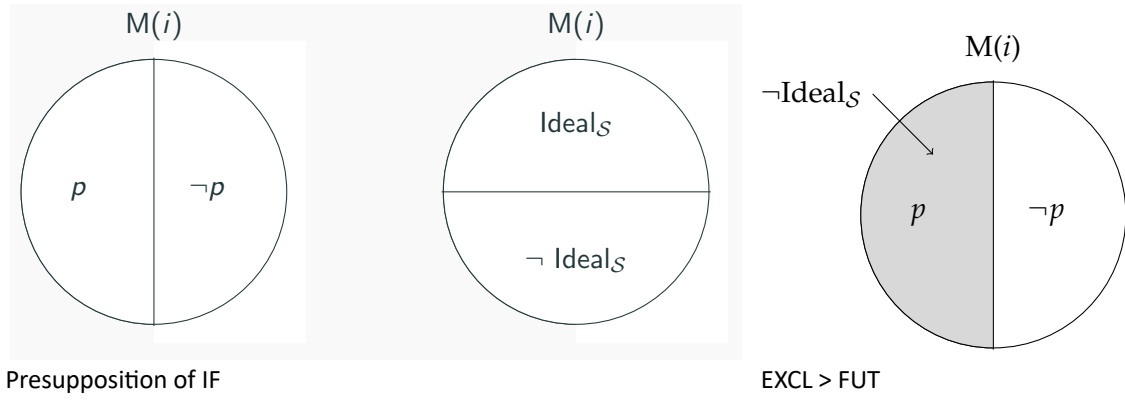


Figure 4: Interpretation of future exclamatives

Epistemic modals such as *potere* (*might*) and *dovere* (*must*), in spite of some degree of flexibility, are less natural than the future with contrariness biases (86).

- (86) a. #Posso essere scema !
 might.1SG be stupid !
 #I might be stupid !
 b. #Devo essere scema !
 must.1SG be stupid !
 #I must be stupid !

However, an in-depth study of their flexibility remains to be undertaken (for new psycholinguistic results, see Liu (2019); Liu and Rotter (2025)), as corpus data reveals that similar uses for epistemic *dovere* (*must*) are indeed attested, raising the new question of the range of negative biases of modal auxiliaries.

- (87) Perché ho accettato tutto questo? Devo essere stupido!
 Why did I accept all this? I must be stupid!²⁷

3 Conclusion

The paper has addressed the understudied question of contrariness biases arising with modals in exclamative questions and modals in exclamatives, and has highlighted two different mechanisms.

In the first case, QUES behaves as expected as an alternative generator, but, failing to producing a felicitous question, it interacts with MICA to return a felicitous one. We have argued that the interplay between the metaevaluative layer of the modal filled by MICA and QUES, generates exclamative questions. In the second case, EXCL acts as an evaluative itself, reversing the expected bias of the modal claim.

Our analysis has assumed that the modal hosts a metaevaluative content as an inherent part of its meaning, and that this evaluative content can be manipulated. It substantiates the idea that QUES and EXCL reveal speaker's intentions (Brentano, 1874) by considering them as bias manipulators, interacting with or spelling out the metaevaluative component of the modal.

In our account, the speech act operators participate to the composition with the modality, without being reducible to a modality themselves. Current research is unveiling their internal structure (Krifka et al., 2023) and their compositional potential (see a.o. Greenberg and

²⁷<https://www.wattpad.com/979151334-un-piccolo-posto-dentro-al-cuore-sherrinford/page/2>

Wolf (2018)) in non-modal environments. We hope that this paper has contributed new useful insights.

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